A Case Study on Continuous Time Pipeline ADCs

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***Abstract*—This work studies a class of continuous-time (CT) analog-to-digital converters (ADCs) suitable for wideband inputs and incorporating inherent anti-aliasing, thereby relaxing the fundamental noise-power tradeoff in a conventional anti-aliasing filter followed by an ADC signal chain. We present the design principles for each block in a three-stage CT pipeline ADC followed by a 7-bit 4× time-interleaved backend asynchronous SAR (BE ASAR) ADC. Each CT pipeline stage employs a coarse ADC-DAC path to create a residue from a delayed form of the input, which is then processed by a residue filter. The BE ASAR ADC is implemented at the schematic level using bootstrapped sampling switches, a custom comparator, clock-generation, and DAC logic blocks. A digital reconstruction filter was used to stitch the data from all the four stages. The three-stage pipeline when paired with an ideal 7-bit quantizer achieves 52-dB SNDR at 20 MHz for a -6dBFs input. The BE ASAR ADC achieves a simulated ENOB of 6.944 bits, SNDR of 43.57-dB, and SFDR of 48.92-dB at 800 MS/s, and maintains stable dynamic performance across the first Nyquist bandwidth. Finally, the three-stage pipeline ADC and BE ASAR ADC are integrated, and time-domain simulations verify that the backend reconstructed output closely tracks the third-stage pipeline residue.**

***Index Terms*—continuous time pipeline (CTP) ADC, anti-alias, filtering, time-interleaved, asynchronous SAR, oversampling**

I. Introduction

Continuous time (CT) analog-to-digital converters (ADCs) have the following advantages over their discrete time (DT) counterparts. Firstly, CT ADCs have inherent anti-aliasing and therefore, act as a filter plus an ADC in a single unit. Next, they are easy to drive and do not require a drive buffer. This reduces the power consumption and distortion introduced by the buffer [1]. Several studies have demonstrated the above advantages through CT $\Delta\Sigma$ Modulators (CTDSM). However, CTDSMs are fundamentally limited by their need for a high oversampling ratio (OSR) to effectively shape the quantization noise out-of-band. A CT Pipeline (CTP) ADC is an emerging architecture that exploits the advantages of a CT ADC while also providing wide bandwidths [1], [2]. It does this by avoiding the feedback loop of the sampled data that is needed in a CTDSM. In this work, we study the CTP ADC implemented in [3] which achieves a 70-dB SNDR with an effective resolution bandwidth (ERBW) of 100 MHz and approximately 60-dB anti-alias filtering in the 1st Nyquist zone. This work is organized as follows. Section II goes through the architectural details of each block in the CTP ADC. Section III describes the circuit design and aspects of implementation on TSMC 16 nm educational PDK. Section IV summarizes key results and Section V concludes the work highlighting key findings during the course of the project.

II. Architectural Details

Figure 1 shows the block diagram of the three-stage CTP followed by a 7-bit 4× time-interleaved asynchronous SAR (ASAR) converter. Each of the three stages is identical and employs a 9-level midtread quantizer to create a coarse sample of the input which then drives a non-return-to-zero (NRZ) DAC. The output of the NRZ DAC is subtracted from a delayed version of the input to generate a residue $r_n(t)$ at each stage, which is then processed by a residue filter. The residue of the third stage is sampled and the sequence $r_3[n]$ is digitized by the backend (BE) ADC. Since each residue filter has a midband gain of 4, the backend quantization noise is ideally attenuated by a factor of 64 when referred to the ADC input. This allows a 7-bit backend ADC to support an overall resolution approaching 13 bits under ideal digital reconstruction and negligible nonidealities. The sampling rate of the pipeline is 800 MHz and the 3-dB bandwidth of the filter is 110 MHz which corresponds to an OSR≈ 4. The digitized sequences from each stage and the BE ADC are processed off-chip by a digital reconstruction filter (DRF) that employs the DT versions of the respective transfer functions as seen by the input and the residues.

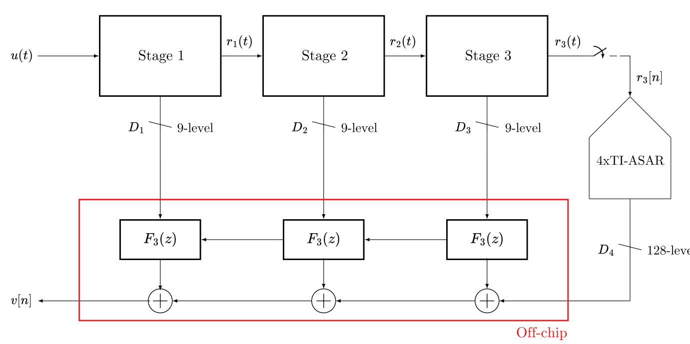

Fig. 1. Architecture of a three-stage CTP with the BE ADC.

A. CT-Pipeline Stages

Figure 2 shows a high-level schematic of a single stage in the pipeline. A 9-level Flash ADC creates a coarse approximation of the input and drives the NRZ DAC. The Flash-DAC path incorporates half a clock cycle delay to allow the comparators to settle. A resistive DAC (RDAC) switches between the appropriate banks of resistors to generate currents corresponding to the ADC output. An RC-lattice network delays the input by the same amount and the residue is generated in current domain.

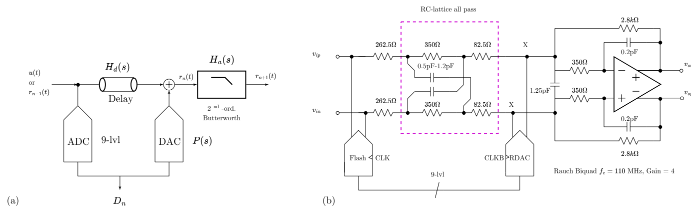

Fig. 2. (a) Block-level diagram of a single stage of the CTP. The input passes through a delay line, an ADC-DAC chain generates the residue, and a residue filter amplifies and filters it. (b) Simplified schematic of each stage. The ADC and DAC are clocked at half a cycle delay.

The residue is then processed by a Rauch biquadratic filter, which realizes a Butterworth low-pass response at a cut-off frequency of 110 MHz and a passband gain of 4. Each stage of the pipeline is identical except for the capacitors in the RC-lattice network which are tuned to adjust the delays.

The DACs in stages 1 and 3 and clocked at CLKB while the DAC in stage 2 is clocked at CLK. The reason is as follows. The first stage's Flash ADC relays its output to the DAC only at the falling edge of the clock. The input is delayed by half a clock cycle through the lattice network, and hence, the residue is now generated from rising edge of CLKB to the next cycle of CLKB. The second stage Flash ADC then starts sampling at the rising edge of CLKB and updates its decision at rising edge of CLK. This logic is extended to further stages. This clocking strategy was accounted for during the digital reconstruction by moving the bit streams by the appropriate number of cycles.

B. Backend 4× Time-Interleaved Asynchronous SAR ADC

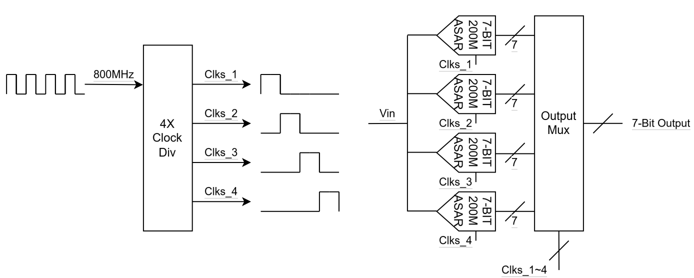

Fig. 3. Architecture of the 4× time-interleaved 7-bit backend ADC.

Figure 3 shows the overall architecture of the 4× time-interleaved backend asynchronous SAR (BE ASAR) ADC. A 4× clock divider generates four-phase 200 MHz clocks for the four ADC slices. The duty cycle and sampling period of each ADC is set to 25%, allowing each ADC slice three 800 MHz clock cycles for conversion while preventing overlap between the input sampling periods of different slices, thereby reducing input-dependent distortion. The output multiplexer digitally selects each ADC output at the corresponding clock rising edge; therefore, each ADC slice completes one conversion within one 200 MHz clock period after rising edge of its sampling phase.

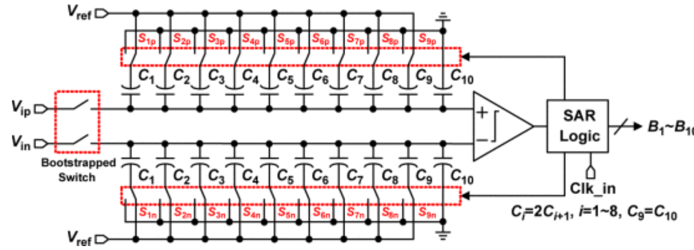

Fig. 4. Architecture of each 7-bit backend ADC slice.

Figure 4 illustrates the architecture of each 7-bit, 200 MHz ASAR ADC slice. The ASAR architecture is adapted from [4], and each ADC slice employs a monotonic switching procedure. The ADC slice itself is fully implemented at the schematic level, whereas the time-interleaving blocks, including the clock divider and output multiplexer, are modeled using Verilog-A.

C. Digital Reconstruction Filter

The principle behind generating the DRF is as follows. In order to reconstruct the input, the coarse approximation needs to go through the same transfer function in discrete time domain as the residue does in continuous time. A residue from stage k passes through the residue filter in its own stage, and the delay lines and residue filters of subsequent stages. Therefore, the coarse bit sequence D_k should go through a transfer function $H_k(z)$ as given by equation 1

$$H_k(z) = \mathrm{c2d}\{P(s)\prod_{i=k}^{n} H_{a,i}(s) \prod_{i=k+1}^{n} H_{d,i}(s)\} \qquad (1)$$

where $P(s)$ is the NRZ DAC's transfer function, and $H_{a,i}(s)$ is the residue filter's and $H_{d,i}(s)$ is the delay line's transfer functions respectively in the i^{th}-stage. The c2d operator in this work implements an impulse invariant transform.

While $H_k(z)$ serves as a good estimate for the reconstruction, it is ideally an infinite impulse response (IIR) filter, and therefore needs to be approximated by a suitable finite impulse response (FIR) filter [2]. A foreground calibration method is discussed in [3] and a modified version of that is implemented in this work. The ADC is excited by a known frequency and the output bit streams $D_1 \cdots D_4$ are recorded. An ideal reference is used to compare the outputs with initial estimates derived from the respective $H_k(z)$ and the weights of three 10-tap FIR filters are tuned so as to minimize the quantization errors in each sample between the reference and reconstructed sequences. These weights are frozen and are used for every subsequent result. In addition to this, the bit streams were also adjusted to account for the pipeline delays.

III. Circuit Design

Figure 2(b) shows the schematic of each stage of the pipeline. The design of the RC-lattice network, the comparators and other design choices in the Flash ADC, the RDAC, and the OTA for the Rauch filter are discussed next followed by the components of the ASAR.

A. RC-Lattice Delay Network

The input is delayed to reduce the peak-to-peak amplitude of the residue. An RC-lattice network as highlighted in Figure 2(b) is implemented to achieve this. Ignoring the source and load resistance, the transfer function from input to output of such a networks shows the following all pass behavior

$$H(s) = \frac{1 - sRC}{1 + sRC} \tag{2}$$

where R and C can be tuned to give the desired delay. Of course, unlike what equation 2 might suggest, there is finite attenuation in the implemented RC-lattice network due to source and load impedances.

B. Flash ADC Design

The 9-level Flash ADC incorporates a conventional double-tail comparator in a differential differencing fashion as shown in Figure 5(a). This comparator choice fits well with the sub-1V V_{DD} supply in this PDK as it reduces stacking of devices as seen in a StrongArm latch, and offers much higher immunity against kickback [5]. The sizing for the comparators was guided through g_m/I_D scripting to sweep the space for device drive strengths and parasitic capacitances that yield a total comparator decision time of less than half clock cycle [6]. The total delay expression was taken from [5]. The results from this modeling are plotted in Figure 5(b) and we see that thanks to the low parasitic capacitances in this node, even minimum sized devices (1-fin, minimum length, and minimum width) comfortably achieve settling much quicker than the target of 625 ps.

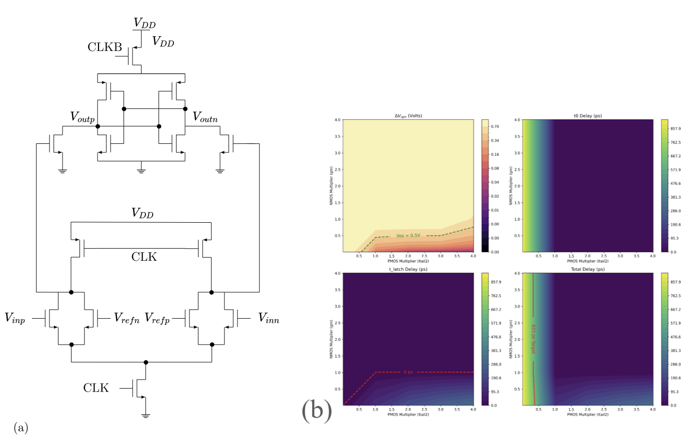

Fig. 5. (a) Schematic of the conventional double-tail comparator used in the Flash ADC. (b) Results of comparator delay modeling as functions of the device sizing.

The decision of the comparator is latched up using a bank of RS-latches and a D flip-flop (DFF) is used to update the output at the falling edge of the clock. These auxiliary blocks were implemented in Verilog-A.

C. RDAC Design

Figure 6 shows the unit element inside the RDAC. The resistance values are chosen to match the resistance value for the Rauch filter network. CMOS switches were sized so as to minimize their R_{on}. Eight such unit elements are necessary to drive the current that covers the full-scale. Since the DFF already updates the output only at the rising edge of CLKB (CLK), the RDAC was not clocked separately.

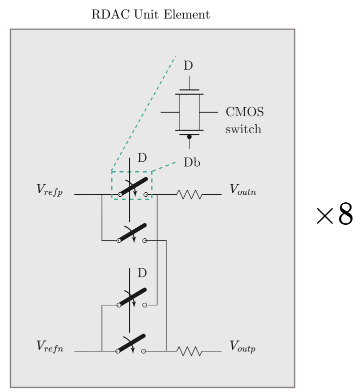

Fig. 6. RDAC unit element is built from reference-switched resistor banks. A CMOS switch is utilized and sized to minimize its on resistance.

D. Filter Design

The passive elements for the Rauch biquad were picked to achieve a maximally flat response at 110 MHz and a passband gain of 4. The OTA for the filter was realized by a three-stage feedforward-compensated topology as shown in Figure 7. With suitable approximations, the transfer function of such a topology is given by equation 3.

$$\frac{v_{out}}{v_{in}} = -\frac{g_{m1}g_{m2}g_{m3}}{G_1G_2G_L}\frac{\left(1+\frac{sg_{m4}C_1}{g_{m1}g_{m2}}\right)\left(1+\frac{sg_{m5}C_2}{g_{m3}g_{m4}}\right)}{\left(1+\frac{sC_1}{G_1}\right)\left(1+\frac{sC_2}{G_2}\right)\left(1+\frac{sC_L}{G_L}\right)} \tag{3}$$

Equation 3 was used to design the necessary poles and zeros to achieve a unity-gain bandwidth (UGBW) in open-loop of 2 GHz. The DC gain target initially was set to 60 dB but was increased to 100 dB in successive design iterations to account for loading from the subsequent stages.

The schematic-level implementation is shown in Figure 8. The first stage utilizes a complementary input stage to double the transconductance, and a cross-coupled NMOS pair to introduce negative resistance and increase its DC gain. This cross-coupled NMOS pair is source-degenerated in the common-mode picture to adjust its V^*. The second stage reuses the bias current to implement both g_{m2} and g_{m4} while the third stage does the same with g_{m3} and g_{m5}. The second

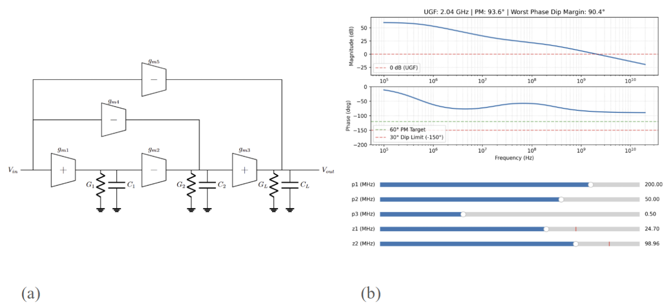

Fig. 7. (a) Block-level implementation of the OTA. (b) Transfer function tuning ensures that we have the poles and zeros at the right location to achieve the desired UGBW and phase margin.

feedforward stage is capacitively coupled so as to bias this stage to allow for maximum output swing. Each stage employs an ideal current-injection common-mode feedback (CMFB) loop that is implemented with ideal vcvs and vccs blocks on Cadence. Since the CMFB loop does not tune the tail biases, the biasing voltages for these devices were also ideal.

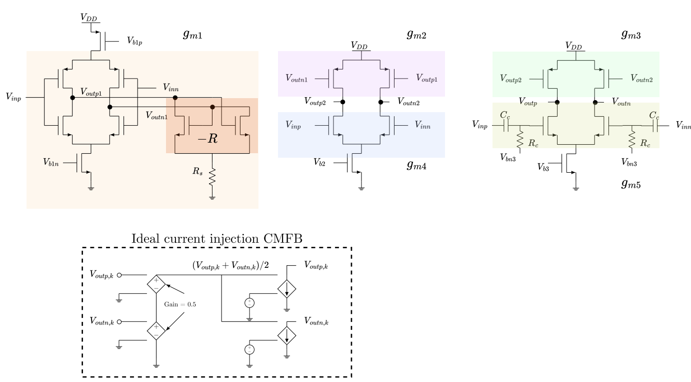

Fig. 8. Schematic-level implementation of the OTA including the ideal CMFB block that was used for each stage.

E. ASAR Bootstrapped Switch

Figure 9(a) shows the schematic of the bootstrapped switch used in the 200 MHz ASAR ADC slice. The on-resistance, R_{on}, of the switch was designed to satisfy the sampling settling requirement of less than 0.01%. The required settling condition is given by

$$\ln(10^4)\tau = 9.2 R_{\text{on}} C_{\text{sample}} < T_{\text{sample}}. \quad (4)$$

From (4), with $C_{\text{sample}} = 0.3$ pF, R_{on} must be lower than approximately 1 kΩ. The switch transistor width was therefore sized to achieve an R_{on} of approximately 200 Ω, providing sufficient margin. In addition, the bootstrapping capacitance, C_s, was set to 50 fF to ensure sufficient gate-source voltage tracking.

F. ASAR Comparator

Figure 9(b) shows the schematic of the comparator used in the 200 MHz ASAR ADC slice. The input transistors were first

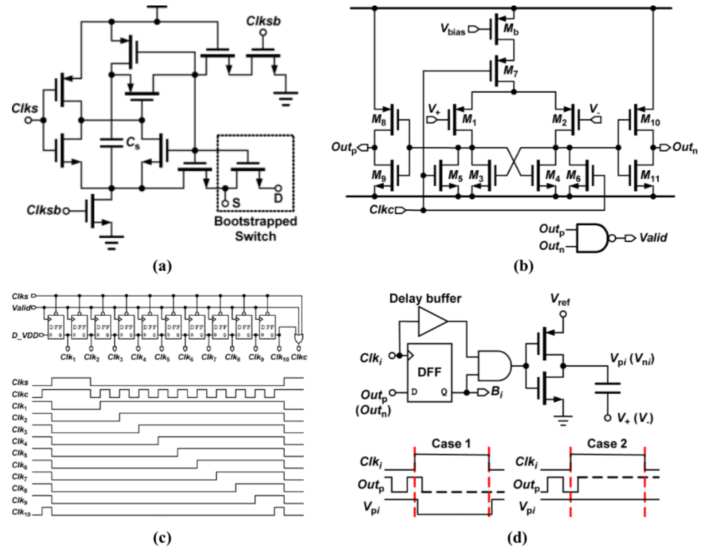

Fig. 9. Internal block schematics of the ASAR ADC: (a) bootstrapped switch, (b) comparator, (c) ASAR internal clock generator, and (d) DAC logic with buffer.

sized to satisfy the mismatch requirement [7]. The logic gates, including transistors M8-M11 and the NAND2 gates, were then sized to achieve the desired transition time. Transistors M3-M6 were sized to provide sufficient pull-down strength. Finally, the bias transistor M_b was sized to provide sufficient pull-up strength by adjusting the bias current.

G. ASAR Clock Generator and DAC Logic

Figure 9(c) and (d) show the schematics of the ASAR internal clock generator and DAC logic used in the 200 MHz ASAR ADC slice. The custom-designed DFFs and logic gates are implemented using minimum-sized transistors, and the cascaded logic paths are optimized to reduce logical effort. The final buffer stage in the DAC logic is sized according to the load capacitance of the corresponding sampling capacitor.

IV. Results

We first discuss the filter characterization and then the results pertaining to the three pipelined stages alone. The BE quantizer for these results was an ideal 7-bit Verilog-A model. The ASAR characterization and results are discussed in detail next. Finally, both the blocks are integrated and preliminary results are shown.

A. OTA and Filter Performance

Figure 10(a) shows the open-loop response of the OTA. The output was loaded down by a 4 pF capacitor to emulate the loading at high frequencies from the next stage's RC-lattice network and the Flash ADC. The DC gain is 98-dB with a UGBW around 1.8 GHz. Figure 10(b) shows the loop-gain of the filter network with the RC-lattice included. Since the feedback network has an additional pole around the target cut-off frequency, we expect to see a UBGW close to it. Note that the resistive load from the lattice network drops the DC gain to around 84 dB. Figure 10(c) shows the filter response and it behaves as expected with a passband gain of 12.5 dB and a cut-off frequency at 108 MHz. Finally, to validate the filter's functionality in the pipeline chain, we analyze the transient output from each stage with the RDAC disconnected. In Figure 10(d), while we expect around 3-dB loss at 100 MHz compared to the 20 MHz signal, the additional 3-db loss is attributed to the delay-tuning done to stages 2 and 3 which increased the capacitive loading on the filter.

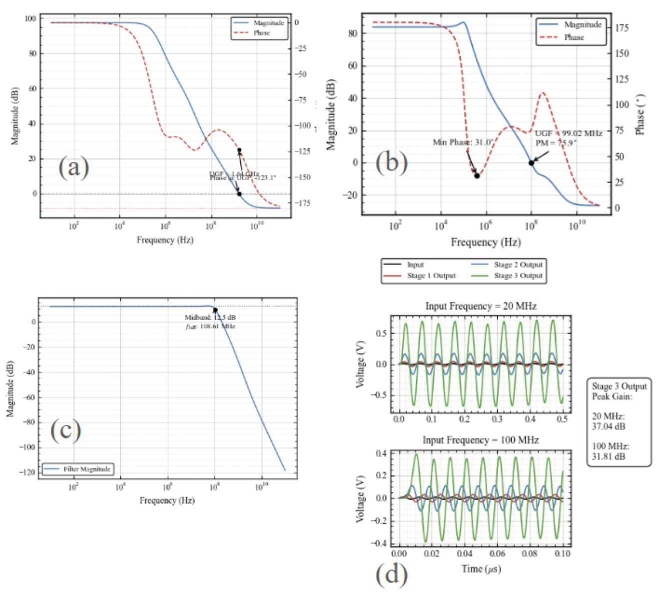

Fig. 10. (a) Open-loop response of the OTA. (b) Loop-gain of the filter network with RC-lattice included. (c) Magnitude Bode plot of the filter response. (d) Time domain response of the filter as connected in the pipeline. The DAC is disconnected to prevent the filter output from being canceled.

B. Pipeline Stages + Ideal 7-bit BE ADC Performance

Figure 11(a) plots the SNDR of the three-stage pipeline with an ideal 7-bit BE quantizer. The peak SNDR achieved for a -6dbFS input is 52-db and corresponds to an ENOB of 9.4-bit when corrected to full-scale. The dynamic range of the pipeline was tested and summarized in Figure 11 (b). The impact of thermal noise on the performance is apparent from Figure 11(c) where we see a loss of 5.7dB as compared to the simulations without transient noise. The PDK was known to have issues with the noise models. We note that even without noise, the errors from the digital reconstruction filter, and other non-idealities from the residue filter limit the peak SNDR to 58dB at 20 MHz. One of the key advantages of the CTP architecture is its inherent anti-aliasing and this is demonstrated in Figure 11(d) where a -20dbFS input tone at 20 MHz was injected, followed by its equivalent in the 1st Nyquist zone. The pipeline provides a rejection of 68dB to the aliased signal.

It is reasonable to question the rapid loss in SNDR vs. frequency as seen in Figure 11(a) despite the filter showing the appropriate transient response at higher frequencies. One

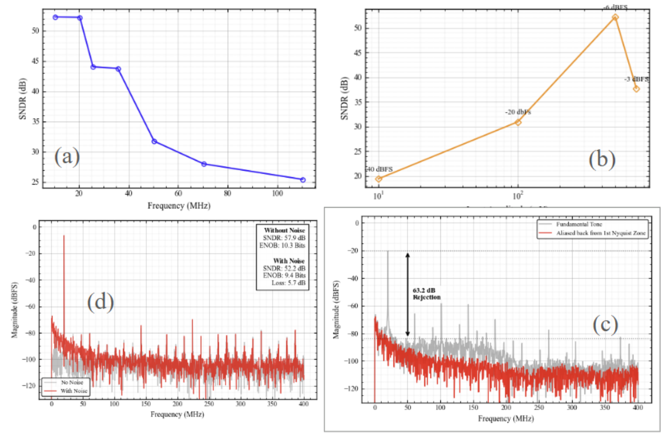

Fig. 11. (a) SNDR vs. frequency for a -6dbFS input. The simulations include transient noise. (b) Dynamic range of the CTP ADC at 20 MHz input. (c) Thermal noise deteriorates the SNDR by almost 1-bit at 20 MHz. (d) The CTP ADC shows greater than 3rd-order anti-alias filtering for a 20 MHz tone from the 1st Nyquist zone.

attempt to fix this was through adjusting the delay in the later stages of the pipeline to reduce the peak residue. However, this only had marginal benefits.

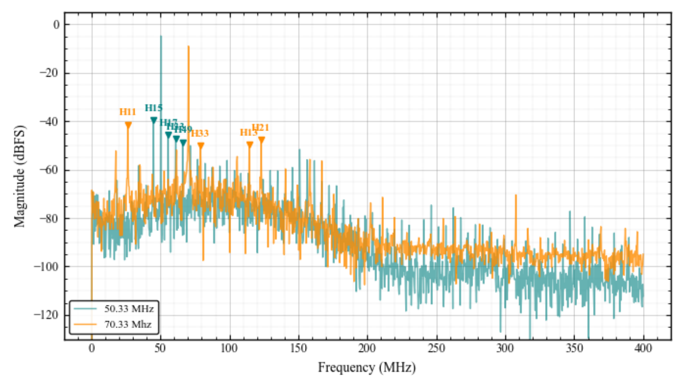

Fig. 12. Higher-order odd harmonics are aliased in-band to create huge spurs and ultimately deteriorate SNDR.

While this issue was not fully explored due to time constraints, a few possible culprits are discussed here. Figure 12 shows the spectrum of a -6dbFS input at 50 MHz and 70 MHz. While the noise floor sits comfortably below -60-dB across the spectrum, there are several spurs that reach up to -40-dB. The first suspect is the robustness of the DRF which is prone to spurs in time domain at every MSB transition if not done right. The least means square algorithm employed here naively tries to minimize the error for each sample versus the ideal reference for the calibration tone. This is slightly different from [3] and could have been reason for the spurs close to the signal frequency in Figure 12. On closer inspection, we see that these are higher-order odd harmonics of the input, such at the 11st, 15st, 21st, etc., are aliased back in-band. We notice that there are no lower-order harmonics that potentially could alias back in-band within the low-pass corner of the residue filter. Despite [3] showing that the DAC images are filtered by the biquad, we hypothesize here that the input waveform is potentially getting mixed with the DAC waveform which contains higher-order odd harmonics with reasonably large energy content which could ultimately be responsible for these spurs. The low-pass filter cannot attenuate these signals if the higher-order harmonics, say 11$^{\text{st}}$-order, has a large energy content, since when it aliases back in-band, it falls well within 110 MHz. The above hypothesis was not, however, rigorously tested. Another potential reason could be the inherent nonlinearity from the OTA [8]. This work does not incorporate the dummy DAC as shown in [3] but we believe this might not explain the entire picture given that we have ideal voltage sources as references. In any case, a thorough investigation is warranted to make any conclusive statements.

C. ASAR Performance

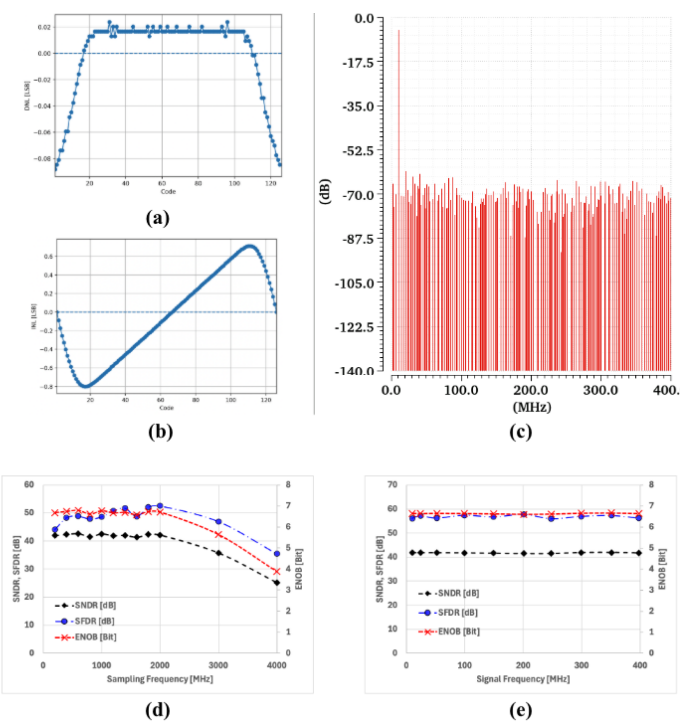

Fig. 13. ASAR ADC simulation results: (a) DNL, (b) INL, (c) 1024-point FFT spectrum at 800 MS/s (d) Performance vs sampling frequency (e) Performance vs signal frequency.

Figure 13(a) and (b) show the simulated DNL and INL performance of the BE ASAR ADC for an input range of -0.7 to 0.7 V. Mismatch and offset effects were not included in this simulation. The simulated minimum/maximum DNL and INL are $-0.088/0.024$ LSB and $-0.80/0.71$ LSB, respectively. The dominant source of the observed DNL/INL degradation is likely comparator nonlinearity at large input amplitudes. When the input range is restricted to -0.5 to 0.5 V, the DNL improves to $-0.004/0.007$ LSB and the INL improves to $-0.005/0.003$ LSB.

Figure 13(c) shows the simulated FFT spectrum for a 10.16 MHz input signal sampled at 800 MS/s. The measured ENOB, SNDR, and SFDR are 6.944 bits, 43.57 dB, and 48.92 dB, respectively.

Figure 13(d) shows the simulated dynamic performance of the ASAR ADC at different sampling frequencies. The ADC maintains nearly constant performance up to 2 GHz, while the performance begins to degrade beyond 3 GHz due to insufficient comparator decision time during conversion.

Figure 13(e) shows the ADC performance at an 800 MS/s sampling rate. The ADC maintains stable performance across the full first Nyquist bandwidth of 400 MHz.

D. Integration

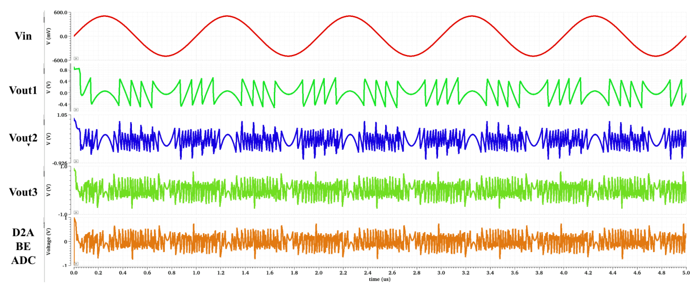

Fig. 14. Integrated 3 stage Pipeline / BE ASAR ADC time-domain output waveform

Figure 14 shows the overall time-domain output waveforms of the integrated three-stage pipeline ADC with the BE ASAR ADC. From top to bottom, the waveforms show the input voltage, the first-, second-, and third-stage pipeline ADC outputs, and the digitally reconstructed analog waveform from the BE ASAR ADC output. A 1-MHz sine wave is applied at the input. The pipeline ADC operates with a 1.0-V supply, while the ASAR ADC operates with a 0.8-V supply. Due to this supply-voltage mismatch, a $0.7\times$ gain block is inserted between the third-stage pipeline ADC output and the BE ASAR ADC input. Each stage output shows the intended transient behavior, and the reconstructed BE ASAR ADC output closely tracks the third-stage pipeline ADC output.

V. Conclusion

In this work, we studied and implemented a three-stage continuous-time pipeline ADC followed by a 7-bit $4\times$ time-interleaved backend ASAR ADC. The architecture combines the inherent anti-aliasing advantage of CT ADCs with the wideband operation enabled by continuous-time pipelining. The main pipeline blocks, including the RC-lattice delay network, Flash ADC, RDAC, Rauch biquadratic residue filter, and digital reconstruction filter, were analyzed and implemented. The OTA for the residue filter was designed using a three-stage feedforward-compensated topology, achieving a simulated open-loop DC gain of 98-dB and a UGBW of approximately 1.8 GHz under capacitive loading. The resulting residue filter achieves the intended low-pass response with a passband gain of 12.5-dB and a cutoff frequency of 108 MHz.

The SNDR for the three-stage pipeline for a 20 MHz -6dBFS input was found to be 52-dB. The anti-aliasing property and dynamic range of the ADC were tested and reported. While the DRF turned out to be a challenge, and perhaps, the reason for the SNDR degradation at high-frequencies, a preliminary investigation of this emerging class of CT ADCs was done end-to-end.

The backend ASAR ADC was designed as a $4\times$ time-interleaved structure operating at an overall sampling rate of 800 MS/s. Each 7-bit ASAR slice was implemented at the schematic level, including the bootstrapped sampling switch, comparator, internal clock generator, and DAC logic. Simulation results show that the BE ASAR ADC achieves DNL/INL of $-0.088/0.024$ LSB and $-0.80/0.71$ LSB over a -0.7 to 0.7 V input range. For a 10.16-MHz input sampled at 800 MS/s, the ADC achieves an ENOB of 6.944 bits, SNDR of 43.57 dB, and SFDR of 48.92 dB. The ADC maintains stable performance up to 2 GHz sampling frequency and across the first Nyquist bandwidth at 800 MS/s.

Finally, the CT pipeline stages and BE ASAR ADC were integrated and verified through time-domain simulation. Due to the supply mismatch between the 1.0-V pipeline ADC and the 0.8-V ASAR ADC, a $0.7\times$ gain block was inserted between the third-stage residue output and the BE ADC input. The integrated simulation confirms that each pipeline stage exhibits the intended transient behavior and that the digitally reconstructed BE ADC output closely tracks the third-stage pipeline output. These results validate the feasibility of the implemented CT pipeline and BE ASAR ADC integration, while future work should focus on completing full dynamic performance evaluation of the integrated ADC, including calibrated digital reconstruction, noise, mismatch, and post-layout effects.

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